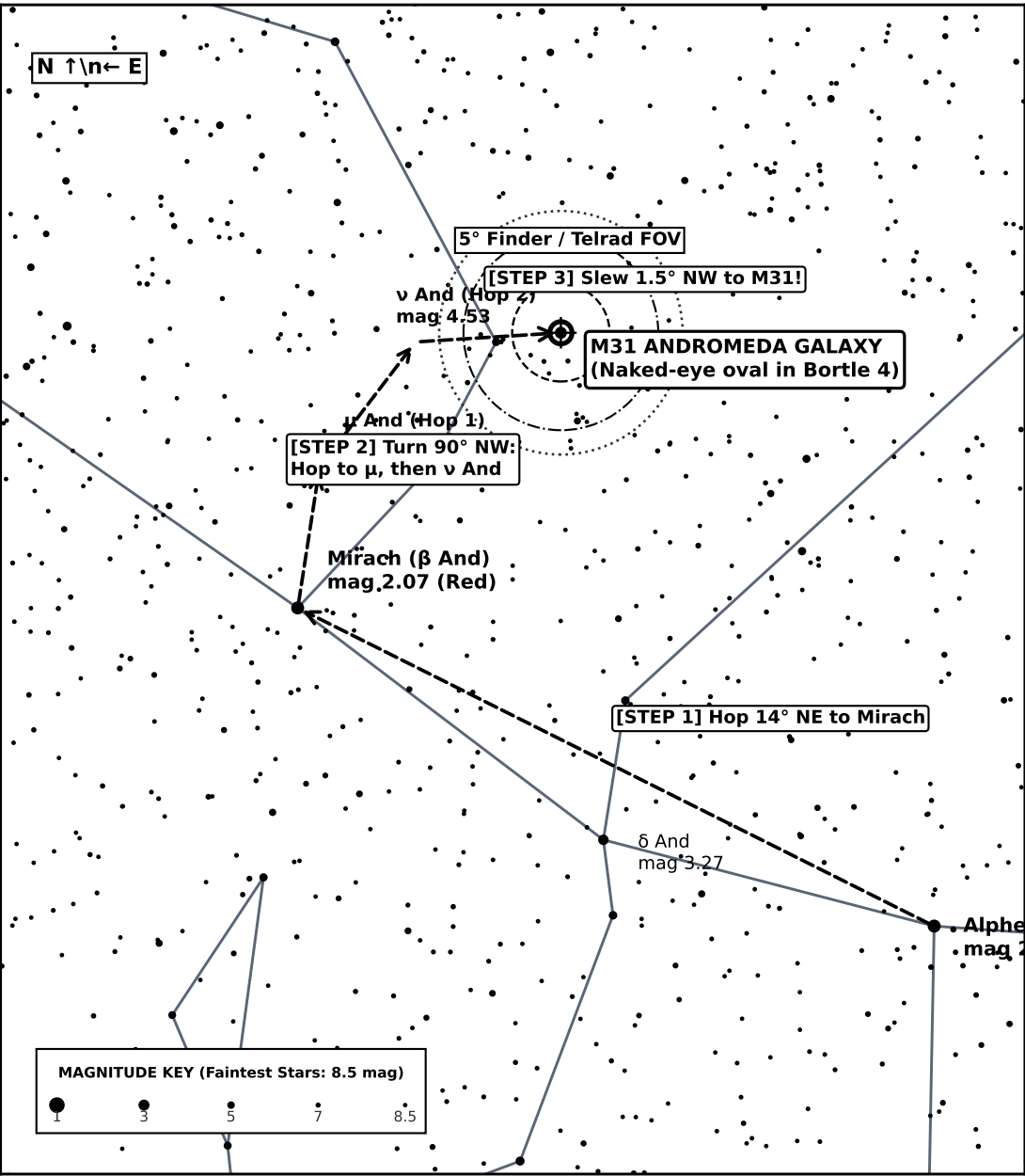


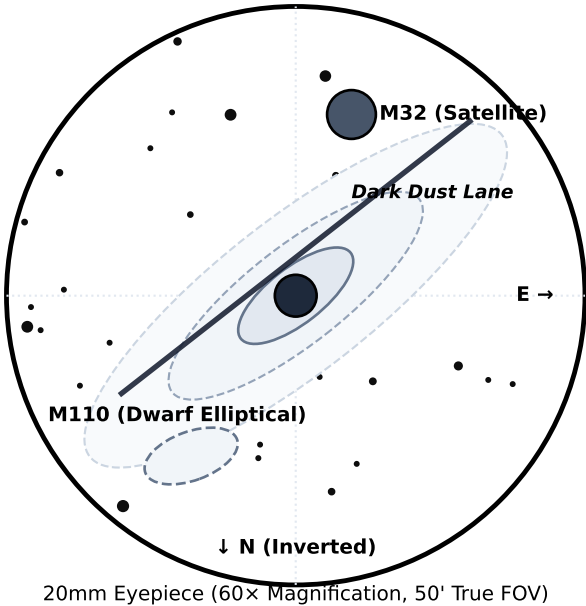
STAR-HOPPING FINDER CHART 4: M31 (ANDROMEDA GALAXY), M32 & M110 — ANDROMEDA

Location: Dvigrad (Bortle 4, SQM 21.0) | Sep 12, 2026 • 21:15 CEST | Sky-Watcher 200P Dobsonian (8-inch f/6, 1200mm FL) | B/W Toner-Saver Edition

VIEWPORT A: WIDE-FIELD STAR-HOPPING CHART (Upright Naked-Eye / Finder: N ↑, E ←)



VIEWPORT B: EYEPIECE SIMULATION (Negative Inverted 180°: N ↓, E →)



VIEWPORT C: TARGET DOSSIER & STAR-HOPPING INSTRUCTIONS

OBJECT DOSSIER: M31 (NGC 224), M32 & M110

- Constellation: Andromeda | Type: Giant Spiral Galaxy
- Magnitude: 3.4V | Surface Brightness: 13.5 | Size: 180' × 63'
- Distance: 2.54 million ly | Companions: M32 & M110
- Tonight's Ephemeris at Dvigrad (Sep 12, 2026):
20:00 CEST: Alt 24.7° | 21:15 CEST: Alt 36.2° (Prime Contrast!)
22:30 CEST: Alt 48.6°, Az 78° (Superb & High)

EYEPIECE QUICK REFERENCE & OBSERVING NOTES:

- Primary Eyepiece (20mm / 60x): Mandatory! Sweeps the immense galactic disk, bright starlike nucleus, and M32.
- Dark Dust Lane: Look outside the core along the NW edge; silhouette dust absorption lane visible with averted vision.
- Companions: M32 (compact south), M110 (diffuse oval NW).

STEP-BY-STEP STAR HOP NARRATIVE:

1. Naked-Eye: Find the Great Square of Pegasus. Identify the northeast corner star Alpheratz (α And).
2. Follow the Andromeda chain: hop 7° to δ And, then 7° to reddish Mirach (β And, mag 2.1).
3. Turn 90° NW: Hop 3.5° to μ And, then 3.5° to ν And.
4. Slew 1.5° further NW: M31 fills the finder and eyepiece!